Coding Assignment 1

Due: 16 October, 2026 6pm PT

Maximum Weight Path on Trees

Problem Description

Consider a tree with N vertices and $N-1$ (undirected) edges, where each edge e is associated with a weight w. Given a path on the tree, its weight is the sum of weights of all the edges in the path.

Please write a program that prints the weight of the path in the tree with maximum weight.

Warning: The input tree may not be a binary tree, so this is not just a regurgitation of a similar algorithm from class.

Input

The input format is as shown below.

N		
u_1	v_1	w_1
u_2	v_2	w_2
...		

More precisely, the input consists of N lines:

- The first line of the input contains one integer N ($4 \leq N \leq 10^5$).
- Each of the next $N-1$ lines describes an edge of the tree with three integers: u_i and v_i, the two endpoints of the edge; and w_i ($-10^4 \leq w_i \leq 10^4$), the length of the edge.

Note: The tree, as given, is unrooted. However, to solve the problem, you will want to root the tree at an arbitrarily chosen vertex. The input is always generated such that, if you root the tree at vertex 1, then every vertex's parent has a smaller index than itself. This will help you avoid messy pre-processing when rooting the tree.

Output

Output 1 line with a single integer, the weight of the maximum weight path in the tree.

Constraints

- $4 \leq N \leq 10^5$
- $1 \leq u_i, v_i \leq N$

- $-10^4 \leq w_i \leq 10^4$
- The time limit is 3 seconds for both C/C++ and Python.

Test Cases

Your program will be evaluated on 5 visible test cases and 5 hidden test cases. Each test case is worth 0.6 points.

The 5 visible test cases are in a single zip file (`CA1.zip`) on Canvas.

Sample Input 1:

```
6
1 2 8
1 3 6
2 4 4
4 5 4
2 6 4
```

Sample Output 1:

```
22
```

Sample Input 2:

```
7
1 2 -5
1 3 11
1 4 -5
2 5 2
5 6 8
2 7 2
```

Sample Output 2:

```
16
```

Sample Explanation

The maximum weight path in Sample 2 is $3-1-2-5-6$, with total weight $11+(-5)+2+8=16$.

Submission Guideline

Write your program in either C, C++ or Python **in a single file**. Name your file `TreePath.ext` where `ext` is `c`, `cpp` or `py` depending on your language. Submit the file on Gradescope. You can make at most 10 submission attempts.

Python Users: Depending on your implementation, your code may need the following two lines:

```
import sys
sys.setrecursionlimit(200000)
```